A 112-Gb/s PAM-4 Long-Reach Wireline Transceiver Using a 36-Way Time-Interleaved SAR ADC and Inverter-Based RX Analog Front-End in 7-nm FinFET

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***Abstract*— A 36-way time-interleaved 56-GS/s 7-bit ADC is designed to realize 112-Gb/s pulse-amplitude modulation (PAM-4) transceiver in a 7-nm FinFET CMOS. The receiver analog front-end stages and the ADC track-and-hold (T/H) buffers are implemented using inverter-based Gm/inverse-Gm-load cells. A distributed inductor peaking network and multi-phase clock calibration is implemented in the quarter-rate transmitter. The transceiver achieves <1E-8 pseudorandom binary sequence (PRBS)-31 PAM-4 bit error rate (BER) over a channel with 37.5-dB loss at 28 GHz while dissipating 602 mW per channel, excluding DSP.**

***Index Terms*— 112 Gb/s, 7-nm FinFET, CMOS, DSP, inverter-based amplifier, pulse-amplitude modulation (PAM-4), time-interleaved SAR ADC, wireline transceiver.**

I. Introduction

INCREASING demand for more bandwidth from data center and communication applications has been accelerating the development of 112-Gb/s transceivers in recent years [1]–[5]. Pulse-amplitude modulation (PAM-4) signaling remains the dominant choice of modulation scheme due to its superior spectral efficiency when compared with nonreturn-to-zero (NRZ) modulation and its simpler implementation when compared with more sophisticated schemes. In many applications, a multitude of these transceivers are monolithically integrated in ASICs and FPGAs to realize high aggregate throughput per chip, therefore mandating the power and area efficiency to be the primary design optimization goal. In addition, these high-throughput VLSIs tend to adopt more advanced process design kits (PDKs) to benefit from the technology scaling that enables compact and energy-efficient implementations of their processing logic cores. While the logic cores benefit from advanced process nodes optimized for digital designs, high-speed transceiver design flows need to commence with a consideration of the constraints and capabilities of a given PDK to result in an efficient solution.

This article reports a long-reach capable 112-Gb/s PAM-4 transceiver designed in a 7-nm FinFET process, extensively using CMOS-style circuit implementations of high bandwidth linear and nonlinear functions needed for data paths and clock paths. Specifically, the entire receiver analog front-end (AFE) and ADC sampling front-end (SFE) are realized by inverter-based gain stages and complementary CMOS pass-gates, obviating the need for current sources and boot-strapped switches. High-speed clock distribution, multi-phase clock generation, duty cycle distortion (DCD)/IQ correction, and phase rotation circuits are also implemented exclusively in the CMOS style. The sub-ADC is implemented in the digital-process-friendly SAR ADC topology. The decision of sub-ADC circuit specification and time-interleaving architecture of the ADC SFE is optimized to fully use the device f_T of the 7-nm FinFET process, to achieve an overall area and power efficiency customized for ultra-high-speed wireline receiver applications.

II. Transceiver Architecture

Fig. 1 illustrates the top-level block diagram of the transceiver, consisted of a shared LC-phase-locked loop (PLL), a transceiver channel, and a control block. Each channel includes a TX, an RX, and a shared channel clock generation block. A shared bias generation and a resistor calibration block are also included. The control block generates a pseudorandom binary sequence (PRBS) pattern, sends 875-MHz 128-b-wide TX data, receives 778-MHz 144-b-wide RX data to and from transceiver channels, and calculates the bit error rate (BER). It includes a raw ADC data capture buffer and interfaces with an off-chip FPGA controller running a background RX adaptation and various calibration loops.

The LC-PLL synthesizes a 14-GHz differential clock from a 583.33-MHz external reference clock. The PLL output clock is

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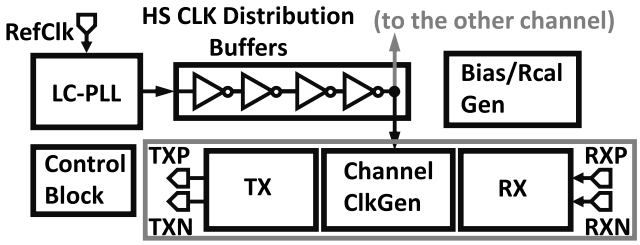


Fig. 1. Transceiver top-level block diagram. (One transceiver lane is shown).

then delivered to each of the two channels via supply-regulated CMOS high-speed clock distribution buffers. The third-order analog LC-PLL uses a compact wide-range transformer-based LC-voltage-controlled oscillator (VCO) [9], optimized for a lower jitter needed for 112-Gb/s operation.

The transceiver design is completed with production specification verified and signed off, including process, voltage, and temperature (PVT) tolerance, timing margin, and reliability checks (electro-migration, device self-heating, bias temperature instability (BTI), HCI, time-dependent dielectric breakdown (TDDB), and electrostatic discharge (ESD), latch-up). Notably, high-bandwidth RX AFE and ADC SFE circuits use thin-oxide MOSFETs for best gain-bandwidth product and f_T available from the 7-nm FinFET PDK. Their device reliability is guaranteed by carefully designing dc bias points both in the normal operation mode and in power down mode such that device terminal voltage differences do not exceed the allowed maximum rating for thin-oxide devices.

III. Transmitter

The TX architecture shown in Fig. 2 adopts a four-tap feed-forward equalizer (FFE) (for two pre-cursor and one post-cursor taps) and a quarter-rate clocking scheme with duty cycle and I/Q mismatch calibration to enable the use of a power-efficient CMOS 4:1 MUX [4]. A current-mode logic (CML) pad driver topology [4], [5] is chosen to achieve 3% finite-impulse response (FIR) resolution without incurring an excessive power penalty from the large number of unit cells containing high-bandwidth node data switches as would be required in a voltage-mode (VM) driver. An array of tail-less CML driver cells with source node switching topology allows the use of more power-efficient CMOS pre-driver and 4:1 MUX. The data path comprises a 128 to 32 serializer, a four-tap FFE data generation, and a 32 to 8 serializer, followed by the final 4:1 MUX. A retimer array is needed to close the timing between the 32:8 serializer and the 4:1 MUX array due to the relatively long routings between the two blocks.

A. Final 4-to-1 MUX

The final 4:1 MUX stage is a performance-critical block in quarter-rate TX designs as it has the tightest timing paths, hence it largely determines the transmitter output jitter performance. Consequently, the block contributes a significant amount to the total TX power consumption. We implemented the 4:1 MUX using a simple CMOS pass-gate-based two-stage MUX clocked by four-phase 14-GHz clocks as shown in Fig. 3. The use of CMOS MUX is motivated by its compact layout and superior energy efficiency.

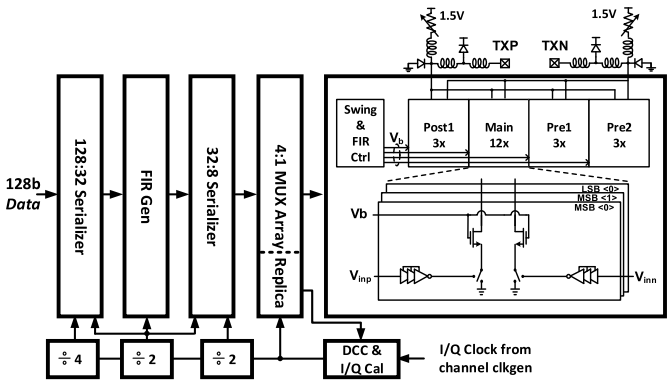


Fig. 2. TX sub-system.

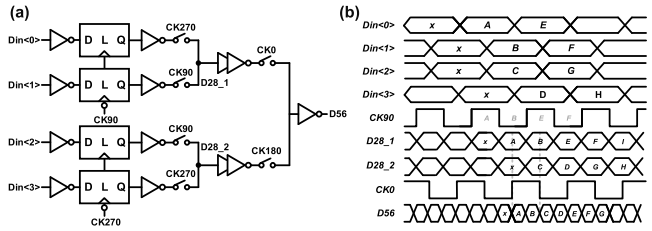


Fig. 3. (a) TX final 4:1 MUX. (b) Timing diagram.

The 14-GHz I/Q clock is produced by the channel clock generation block shared by TX and RX and distributed to TX in full CMOS swing via a regulated CMOS buffer chain. A TX clock calibration block for duty cycle correction (DCC) and I/Q error compensation is needed to correct the duty cycle distortion and I/Q phase errors caused by device mismatches from the MUX circuit so that a low deterministic jitter can be achieved at 112 Gb/s. The phase errors are sensed at the output of the replica MUXs and are background calibrated with smaller than 100-fs resolutions.

B. Pad Driver

A tail-less CML driver [5] is chosen in this design to achieve four-tap FIR (two pre-cursors and one post-cursor) with less than 3% step size. The CML driver is known to consume 4 times more current than the VM driver for the same output swing. However, a CML driver can attain a fine FIR resolution more easily by controlling its DC biasing without affecting high-speed signal paths. In contrast, a slice-based VM driver [4] implements the FIR by configuring the input data of each unit slice, which limits the minimum number of required 4:1 MUXes. Consequently, the amount of power saved in the VM pad driver is not compensated enough for the power overhead of the 4 to 1 MUX and clocking circuits in a quarter-rate architecture with a fine FIR resolution.

In this work, a distributed peaking network is designed to compensate for over 200-fF device and parasitic capacitance at the driver current summing node. A T-coil with an additional series inductor effectively separates the N- and P-type ESD diodes, C4 bump pad, and on-die termination (ODT) resistor array, implementing a lumped transmission line for bandwidth extension and improved return loss. With the distributed peaking network, a 46-GHz bandwidth is achieved at the TX output.

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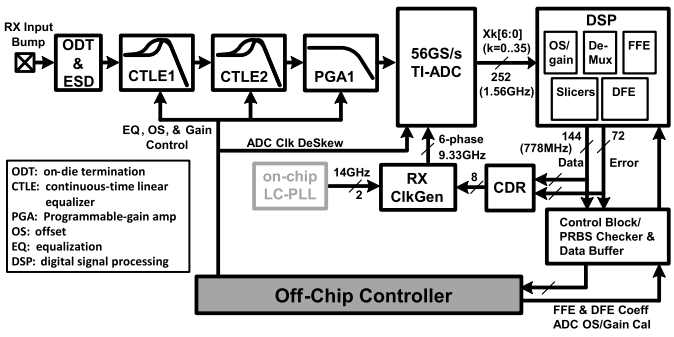


Fig. 4. RX sub-system.

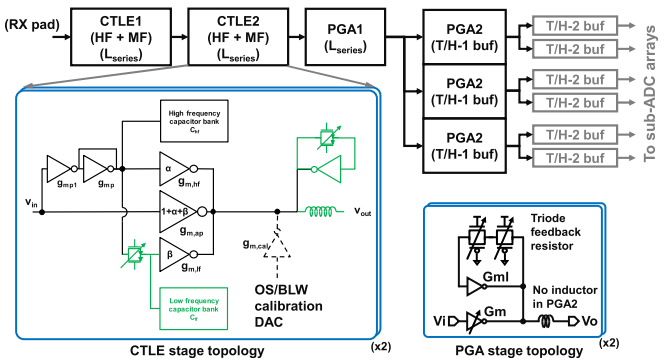


Fig. 5. RX analog front-end implemented using inverter-based Gm/Gml stages.

IV. Receiver

Fig. 4 depicts the RX top-level functional diagram. The PAM-4 signal received at the RXP/RXN input pads is terminated by calibrated ODT resistors. T-coil inductors are used for impedance matching to improve return loss and bandwidth by neutralizing parasitic capacitances caused by input pads, ESD diodes, ODT, and continuous-time linear equalizer (CTLE) input capacitance. The all CMOS RX AFE, consisting of two stages of CTLE, equalizes the input signals by boosting the Nyquist frequency (HF) and mid-range frequency (MF) bands separately. The following two stages of the programmable-gain amplifier (PGA) adjust aggregated broadband gain for automatic gain control adaptation that controls the output signal range for the subsequent ADC to digitize with fully used full scale range (FSR).

The ADC SFE realizes analog signal de-muxing by two track-and-hold (T/H) stages. The second PGA stage acts as the first-rank T/H buffer. A two-rank 6-by-6 time interleaving serially breaks down the ADC input signal into 36 sub-ADCs, which subsequently digitize the signal into 7-b symbols, xk[6:0] ($k = 0..35$). The 6×6 interleaver architecture is chosen with considerations of tradeoffs among the SFE, clocking circuitry, and sub-ADCs' sampling speeds, which will be discussed in further detail.

The DSP retimes ADC output symbols, xk[6:0] ($k = 0..35$), and performs offset and gain correction for the 36 individual sub-ADCs. The 72 deserialized xk are fed into subsequent 31-tap FFE and one-tap speculative decision-feedback equalizer (DFE) stages for ISI cancellation before being converted to produce data and error bits.

An on-chip baud-rate clock and data recovery (CDR) logic detects the optimal ADC sampling aperture phase (T/H-1 clock falling edge) by accumulating the Mueller–Muller [8] phase detector output to update an 8-b phase interpolator (PI) code.

A. Data Path

The RX AFE in Fig. 5 consists of a two-stage CTLE followed by two PGA stages. The CTLEs implement high-frequency peaking with a feedforward pole-subtractive topology, and hybrid peaking (i.e., both HF and MF) is realized in both the stages [8]. At the max peaking setting, the CTLE provides more than 16 dB of peaking at the Nyquist frequency. MF tuning provides additional long-tail ISI cancellation, which helps reduce the post-cursor FFE coefficient ranges. To further reduce the area of the device capacitors for MF path, a series triode device is inserted to allow resistance tuning for the zero frequency. A combination of active-inductor Gm load (Gml) and passive inductor effectively boosts the bandwidth of each stage of CTLE and PGA1. Compact inverter-based amplifier stages consisted of a Gm cell and inverse-Gml, with inherently good linearity due to ratio-metric design [8], are used to implement CTLE, PGA, and ADC T/H buffers. The active inductor is realized by complementary pass-gates in the triode mode, which are used as feedback resistors instead of the conventional TiN resistors, dramatically reducing the cell size and parasitic wire loading. All AFE circuit elements (Gm, Gml, triode-mode feedback resistors, zero resistors, and zero capacitors) are laid out using the same minimum design rule poly-to-poly pitch to achieve high area efficiency and more process uniformity. An overall 70% area reduction (passive inductor area not included) is achieved compared with [8] for each CTLE stage.

The time-interleaved ADC architecture is carefully chosen to optimize the overall power and area efficiency under various constraints of the design. Due to the CMOS nature of the AFE and ADC, the fan-out of each can be decided in a "digital" fashion. Once the design of the sub-ADC is optimized, its input capacitance propagates eventually back to the RX input, determining along the way the drive strengths of the preceding stage given their respective bandwidth requirements. Therefore, we first evaluate the number of required sub-ADCs to minimize fan-out and reduce the overhead incurred by layout complexity and parasitics. To fully take advantage of technology scaling provided by the 7-nm FinFET, the sub-ADC is optimized for sampling speed and area.

The metastability impact on the sub-ADC performance could become the gating factor of the overall system performance. Therefore, an in-depth study was done to understand the effect of metastability-induced errors in asynchronous SAR ADCs [10]. While pushing the sub-ADCs to their speed limit can significantly increase the probabilities of metastability events, the generated errors still could be tolerable in SerDes applications. An intuitive understanding of the detailed analysis in [10] could be established with the following metastable conditions. In a high-speed asynchronous SAR ADC, it is more probable for the LSB bit position to run out of time for evaluation creating an LSB metastable event, but the error created is small (i.e., 0 or 1 LSB code error) and

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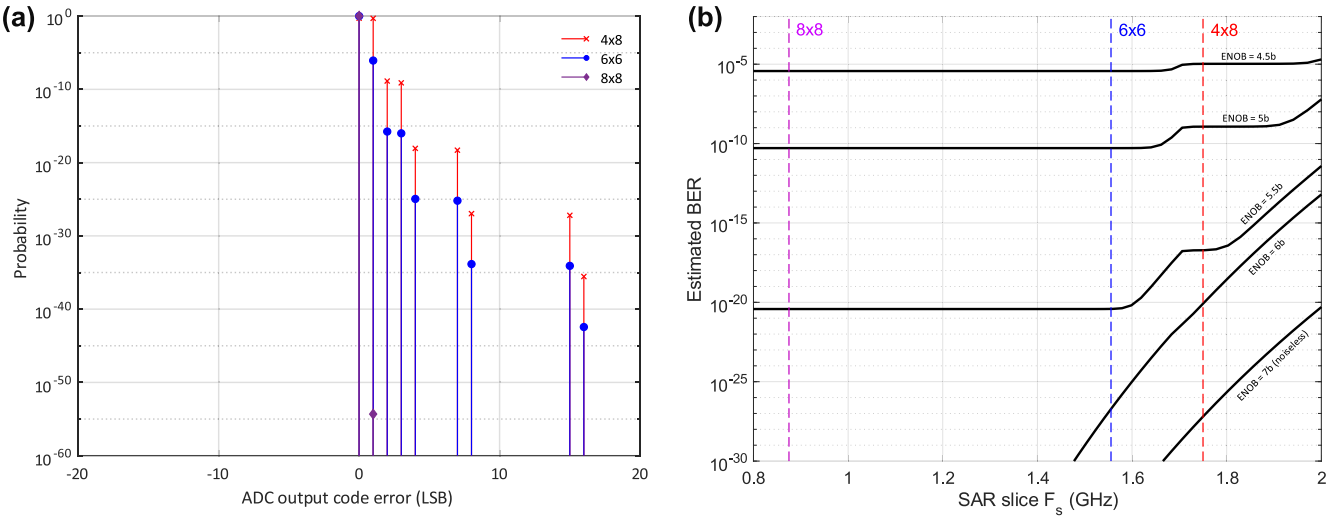


Fig. 6. (a) Simulated 7-b SAR ADC metastability error probability versus error magnitude for 4 × 8, 6 × 6, and 8 × 8 time-interleaving. (b) Simulated system BER versus sampling speed for five cases of thermal noise level.

typically below the thermal noise floor, thus having minimal impact on the BER of the overall system. On the other hand, an MSB bit position metastability event could yield much larger code errors, which almost certainly causes bit/symbol errors. However, these are exponentially unlikely events by construction of asynchronous SAR ADCs. Taking the 7-nm process capabilities and metastability impact on the system BER into account, we concluded that the sub-ADC design in this work can operate at around 1.6 GS/s across different PVT corners with a sufficient margin.

As an illustration, three possible time-interleaving architectures for 56 GS/s are considered for BER impact simulation: 4 × 8, 6 × 6, and 8 × 8, corresponding to sub-ADC sampling speed (F_S) of 1.75 GS/s, 1.56 GS/s, and F_S = 875 MS/s, respectively. Fig. 6(a) shows the simulated metastability error probability versus error magnitude in the unit of LSB for the three cases. The 4 × 8 case shows 1 LSB magnitude error probability almost at the same level as zero error, implying that the LSB bit resolution during the SAR loop almost always runs out of time. Also, 2 to 4 LSB error probabilities are high enough (∼1E-8) to impact on the system BER. On the other hand, the 8 × 8 case shows almost no metastability code errors with less than 1E-50 chance of failing with one LSB error. The 6 × 6 architecture achieves the balance between metastability error probabilities and pushing sub-ADC speed. The one LSB error probability is much smaller compared with that caused by the thermal noise, while higher impact 2 to 4 LSB errors are sufficiently rare (less than 1E-15 chance). Fig. 6(b) shows the simulated system BER versus F_S for five cases of system noise level as modeled by equivalent ENOB ranging from 4.5 to 7 b (noise-less). Three vertical dashed lines mark F_S corresponding to 8 × 8, 6 × 6, and 4 × 8 interleaving architectures, respectively. For 8 × 8, the system BER is always thermal noise limited (i.e., independent from F_S variation) and each SAR ADC slice is far away from their speed capabilities, under-utilizing 7-nm device speed limit. For the other extreme of 4 × 8 architecture, the metastability errors cause significant degradations in system performance, especially at lower baseline BER. For 6 × 6, we are operating closer to the metastability BER corners while the system performance is still dominated by thermal noise.

Given the suitable sampling speed of sub-ADC, various interleave architecture candidates are compared in Fig. 7. The 4 × 8 architecture stresses the sampling bandwidth of the front-end, clocking, and sub-ADC sampling speed despite its area advantage. An 8 × 8 architecture [2] significantly relaxes the speed requirement of each ADC component, but in turn wastes the extra speed offered by the process technology (e.g., concluded 1.6 GS/s capability of the sub-ADC with respect to the required 875-MS/s sampling speed). The area penalty and the associated floor-planning challenge of the 8 × 8 interleaving are also non-negligible due to a larger number of sub-ADC and clock generation. Therefore, the 6 × 6 architecture serves as a good trade-off between effective utilization of each key block design without over-stressing design margins and layout complexity. The performance-critical rank 1 T/H will sample at 9.333 GHz, which is a reasonable frequency for both the analog amplifiers and clock generation and distribution.

Fig. 8(b) illustrates detailed schematics of the 6 × 6 SFE. The SFE requires 3× rank 1 T/H buffers, each sampling with ∼50% non-overlapping clocks. The T/H-1 buffers also act as second-stage PGAs, which are a scaled down version of the inductor-less PGA1. Each T/H-1 buffer drives 2× rank 2 T/H unity gain buffers. All SFE stages use the same inverter-based circuits (Gm and active inductor triodes) as the CTLEs. The output common-mode of the Gm/Gml cells are compatible with the complementary CMOS pass-gates used for all T/H switches in the ADC, obviating the need for bootstrapping circuits and significantly reducing area and clock power consumption. A correct bias voltage is needed at the interface of rank 1 and rank 2 T/H buffers due to the nature of inverter-based T/H-2 Gm cells, and thus clock/charge injection switches are added for the T/H-1 switches. Since the StrongArm sense-amp comparator in

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	4x8	8x8	6x6 [this work]
Total # of sub-ADCs	32	64	36
Sub-ADC sampling speed	1.75GS/s	875MS/s	1.56GS/s
Rank1 sampler frequency	14GHz	7GHz	9.33GHz
Rank1 settling BW (7b)	~28GHz	~13GHz	~18GHz
# of T/H amplifiers	2 + 4	4 + 8	3 + 6
Area penalty	Small	Large	Medium
Clock distribution	Easy	Difficult	Moderate
Challenges	❑ Sub-ADC & rank1 speed ❑ Rank1 settling BW requirement	❑ Area & power penalty ❑ Clock distribution	❑ Manageable area & power penalty (compared to 4x8)
Advantages	❑ Compact layout ❑ Easy clocking	❑ Relaxed BW & timing requirement	❑ Good trade-off between process f_T & design margin

Fig. 7. Design tradeoffs for 4 × 8, 8 × 8, and 6 × 6 time-interleaving architectures for ADC SFE.

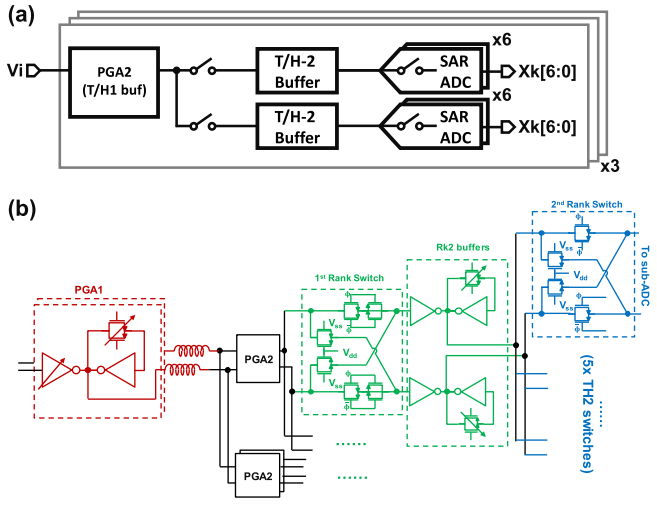


Fig. 8. (a) ADC sub-system and (b) circuit implementation of its sampling front-end.

sub-ADC can operate with a slightly wider common-mode range, the charge injection cancellation devices of T/H-2 switches are removed to reduce the parasitic capacitance at the capacitive digital-to-analog converter (CDAC) node. Both the stages have dummy cross-coupling devices for >50 dB hold mode feedthrough cancellation. The sub-ADC adopted a speed-optimized asynchronous SAR ADC architecture using a binary CDAC with an estimated unit capacitance of 0.5 fF.

B. Clock Generation

The RX clock generation architecture is driven from the top down by some key system constraints and optimizations. The use of a quarter rate architecture in the transmitter gives rise to the need for non-rotating 14-GHz quadrature input clocks, along with an LC PLL running at this frequency. The tradeoffs outlined regarding the speed of each sub-ADC and the number of sub-ADCs to optimize for area, power, and bandwidth result in the RX requirement for six equally spaced phases at 9.333 GHz, with optional phase rotation to track frequency differences between an external TX and the on-chip PLL. Generating the RX clocks from the same clock source as the TX obviates a need for a second clock source. The time-stamp information for a 9.333-GHz clock is already present in a 14-GHz clock, provided the duty cycle is accurate.

The architecture is given in Fig. 9. The 14-GHz clocks from the PLL are used as the injection for a four-stage CMOS 1:1 injection-locked oscillator (ILO). The four-stage ILO uses XOR-based phase sensing with supply regulation [7]. The outputs of this ILO drive both the TX and RX. At the RX, the eight equally spaced output phases at 14 GHz are used to drive an all CMOS 8-bit PI producing a rotating 14-GHz output. This differential clock has variable duty cycle control. The DCD-corrected output provides the source for an IQ-divide-by-6 circuit. The division by 6 is a cascade of a divide-by-3 and a divide-by-2, with one each in parallel for the I- and Q-paths. The quadrature is achieved by releasing the I-path first, followed by the Q-path 1.5 cycles later. The four-phase output at 2.333 GHz contains fewer transitions than a differential 14-GHz clock, thus being advantageous from a distribution power perspective. It is important to note that any DCD in the output of the PI will translate into the IQ error at the output of the divide-by-6 block. As such, IQ correction of the 2.333-GHz clocks is carried out by adjusting the duty cycle after the PI. An analog-style XOR detection is used to generate an error voltage which after processing with a dynamic comparator is used as an input to a digital control algorithm as illustrated in Fig. 9(c).

As illustrated in Fig. 10, the 2.333-GHz IQ clocks are now progressively processed to realize a multi-phase 9.333-GHz clock. The first step is a 4-to-1 self-timed multiplexer used to cyclically select the four 2.333-GHz clocks. This creates a 9.333-GHz pulse train with well-defined low jitter positive edges, but poorly controlled duty cycle. Passing this signal through a pulse generation circuit with adjustable pulse width, as shown in Fig. 10(a), generates a signal suitable for injecting into a ring oscillator pre-calibrated to near

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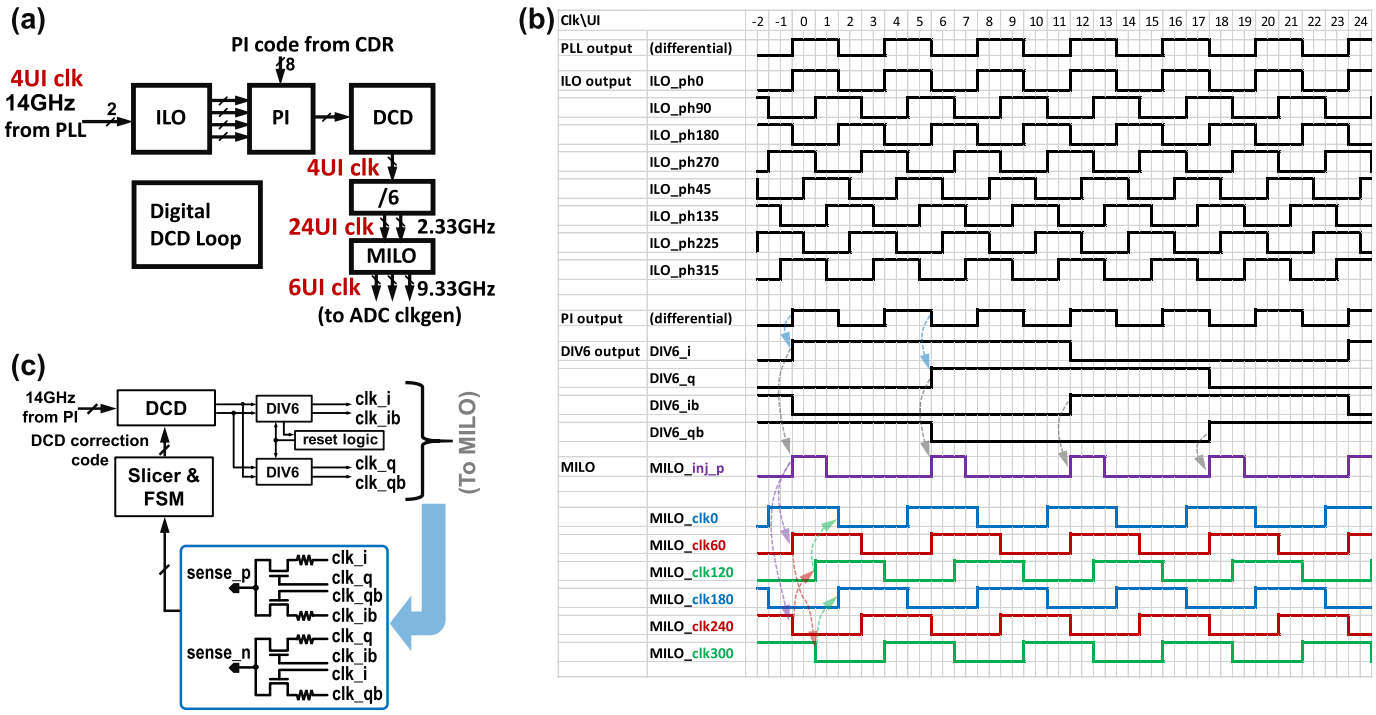


Fig. 9. (a) RX clock generation sub-system. (b) Timing diagram (only cycle-wise accurate). (c) DIV6 and DCD correction loop.

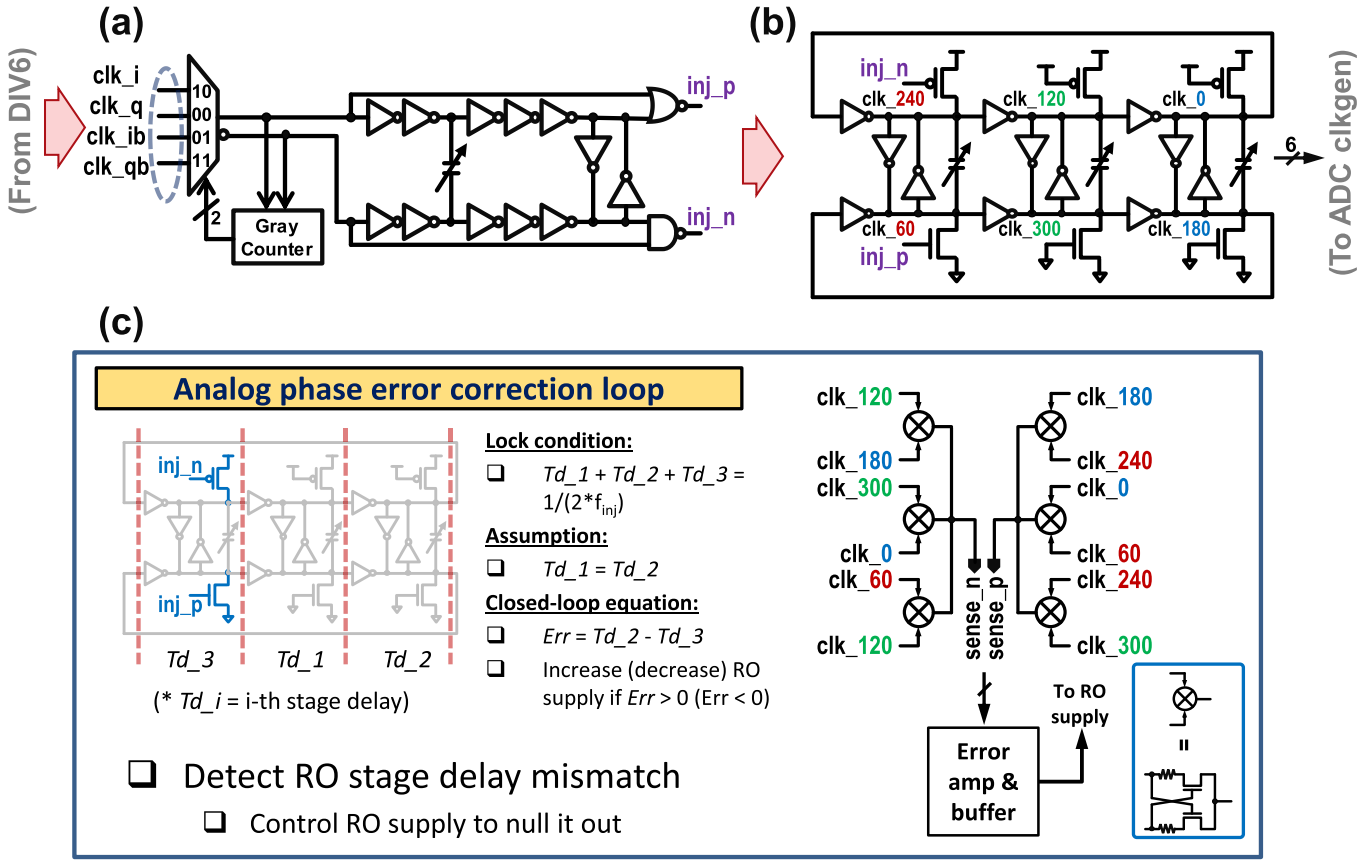


Fig. 10. Multiplying ILO circuits. (a) Pulse generator. (b) Injection-locked ring oscillator. (c) Phase error detector and correction loop.

9.333 GHz [Fig. 10(b)]. The combination of the multiplexer, pulse generator, and tuned ring oscillator together forms a multiplying-ILO (MILO) [11].

Fine control of the natural oscillation frequency of the MILO is required to achieve low phase spacing error at the six-phase output. The supply to the MILO is regulated by

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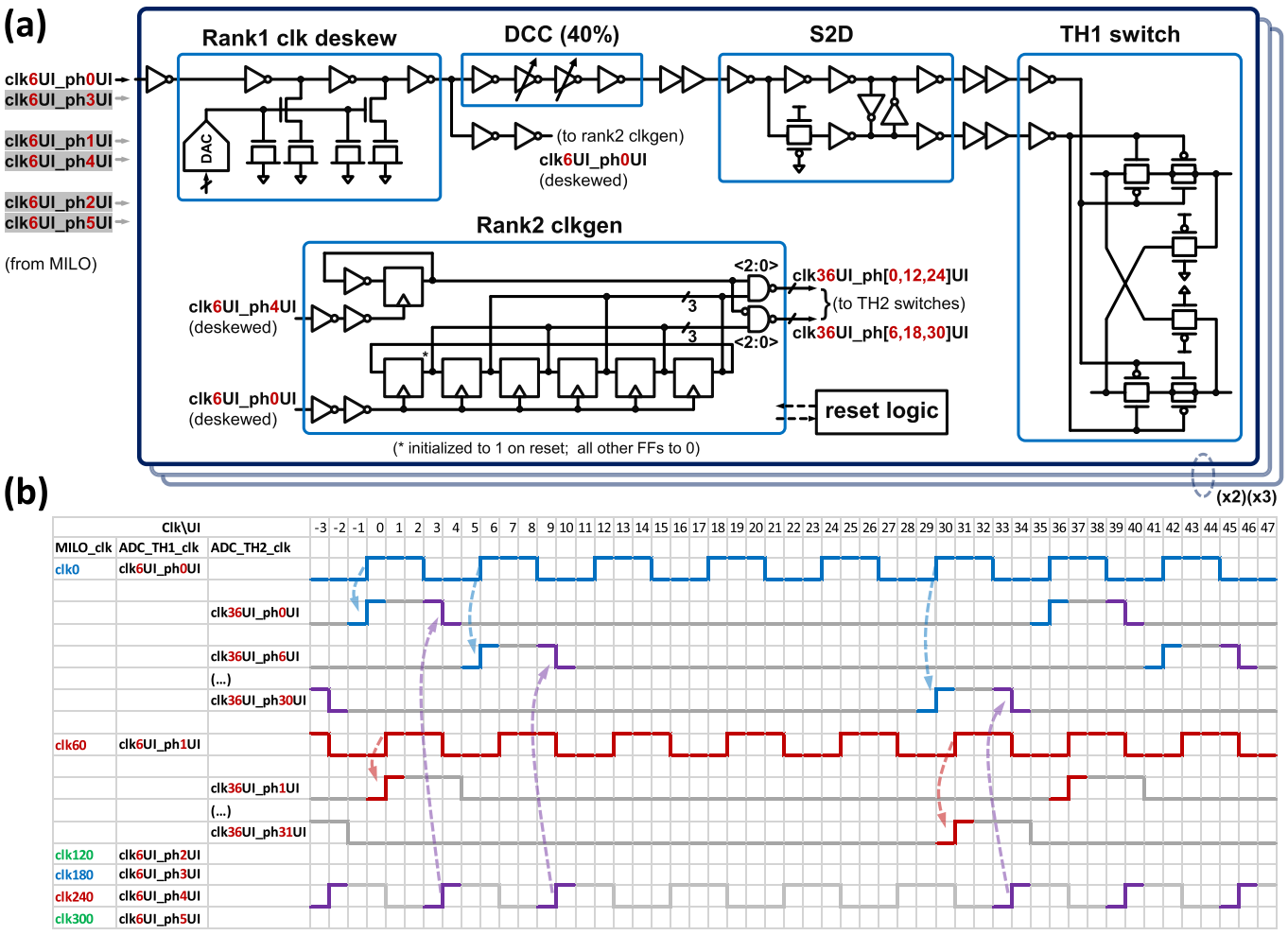


Fig. 11. ADC clock generation for rank 1 and rank 2 clocks. (a) Circuit implementation. (b) Timing diagram.

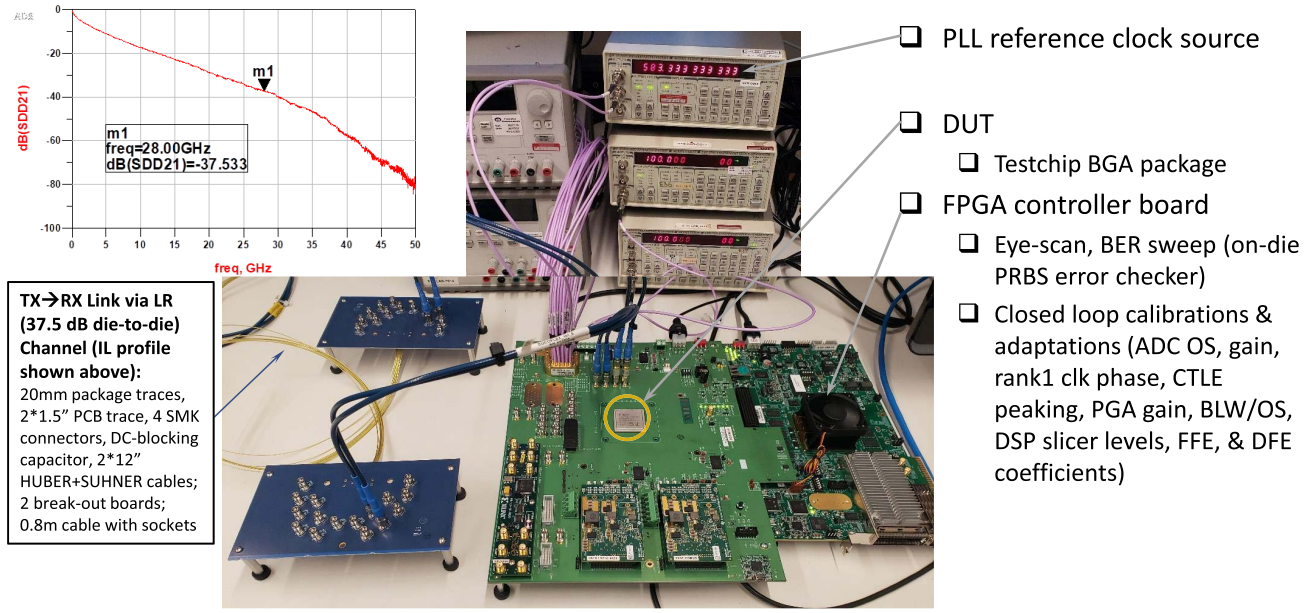


Fig. 12. Bench setup shown with TX to RX long-reach link performance test.

combining the six output phases to produce an error signal which is then integrated and filtered as illustrated in Fig. 10(c). The regulation has the effect of adjusting the natural oscillation frequency of the ILO such that its output period matches the input period. It is important to minimize the offset errors at the phase combination and error amplifier blocks as not all errors

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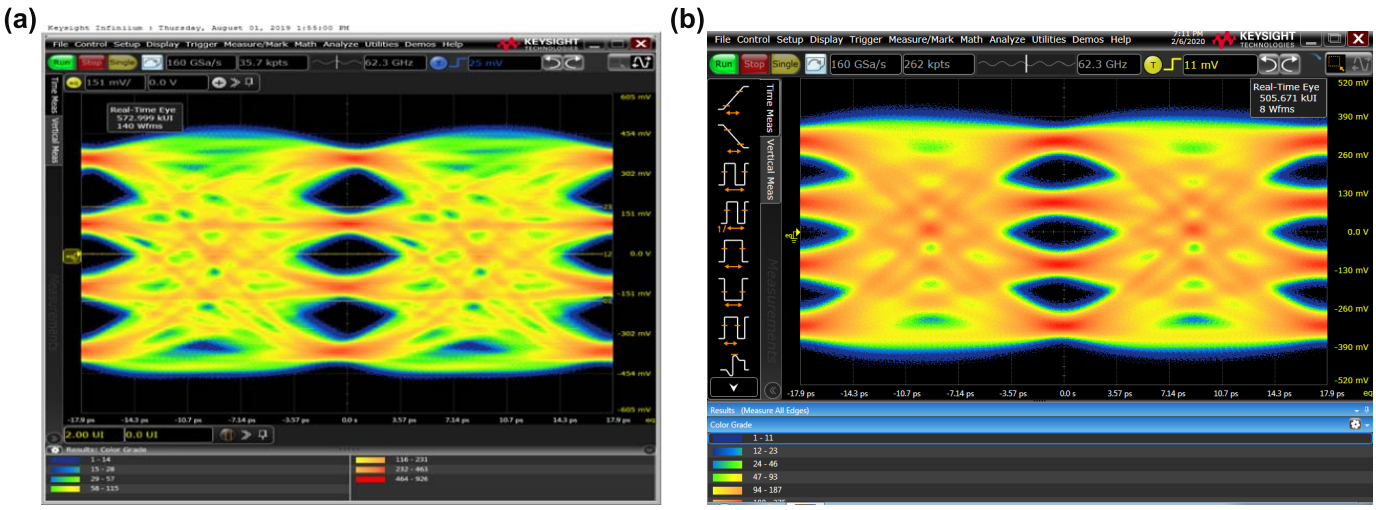


Fig. 13. 112-Gb/s PAM-4 TX output eye diagrams of (a) PRBS-7 and (b) PRBS-31 patterns.

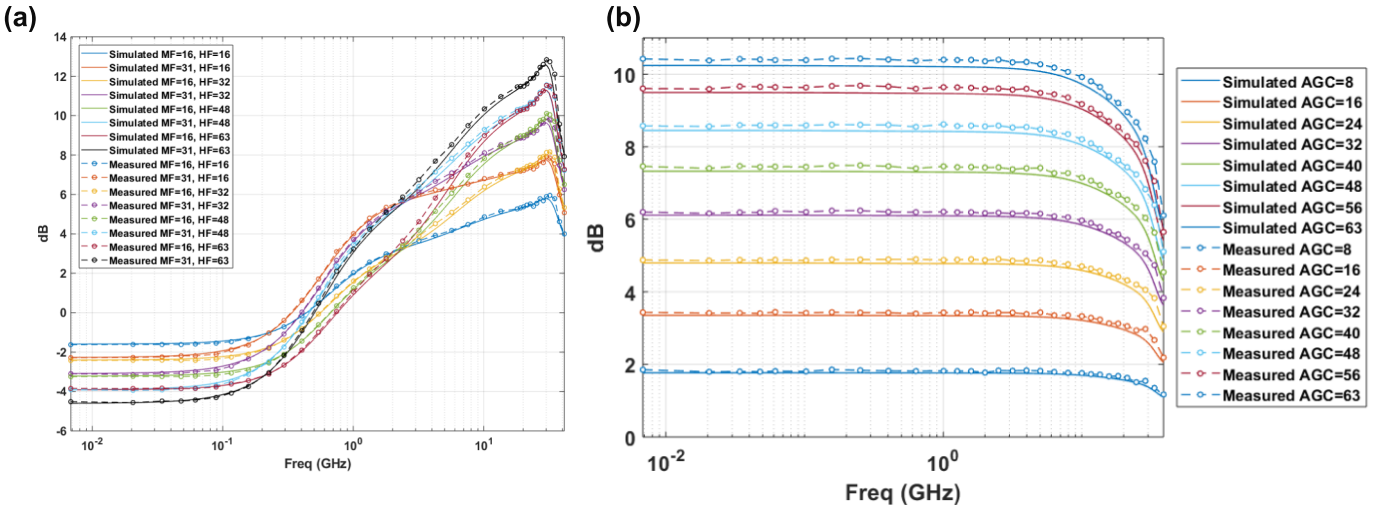


Fig. 14. RX AFE transfer function measurement versus simulation for (a) CTLE and (b) PGA.

accumulated are static or correctable. With a single control point, the average phase error at the MILO is minimized, but care is taken with the physical design to equalize the loading on the ring oscillator nodes to reduce the residual mismatch-induced errors.

The MILO method has the advantage that multi-phase clocks can be generated close to the locality of final use, lowering the distribution power. It should be noted for jitter purposes, however, that while the injected phases can have low-frequency jitter approaching that of the reference clock, high-frequency jitter (beyond the ILO bandwidth) will tend toward that of the intrinsic ring oscillator, and the non-injected phases will have further added jitter. Additional power-supply-induced jitter also requires careful budgeting.

It can be noted that instances of the pulse generation circuit being driven directly by the 2.333-GHz clocks would also produce a series of useful pulses, but that in such a case, multiple injection devices would be required. Instead, the use of fewer but higher rate injection devices lowers the loading on the ring oscillator and reduces multi-path mismatch.

While the MILO does have inherent suppression of timing errors at the injection point, residual offsets and random variations in both the MILO itself and downstream processing require further calibration. As such, the clocks are corrected for both skew and duty cycle, prior to application to the T/H switches in rank 1 or rank 2 ADC time interleavers.

The ADC clock generation circuit in Fig. 11(a) receives 6UI period six phase clocks from the MILO and processes them to produce T/H-1 and T/H-2 switching clocks. The skew correction is carried out using a DAC to adjust an MOS-based load on the clock paths. This affects both the rising and falling clock edges. The skew-adjusted clocks with near 50% duty cycle are divided for use in the rank 2 clocks, while the rank 1 clocks undergo programmable DCD adjustment to achieve a 40% duty cycle (2.4 UI tracking window) at 56GS/s ADC sampling speed. For power efficiency, the skew correction and DCD are applied to the six phases prior to a set of single to differential stages driving the T&H switches. T/H-2 clocks are generated by dividing the skew-corrected 50% duty cycle T/H-1 clocks by 6 to generate six-phase 6UI-long pulses

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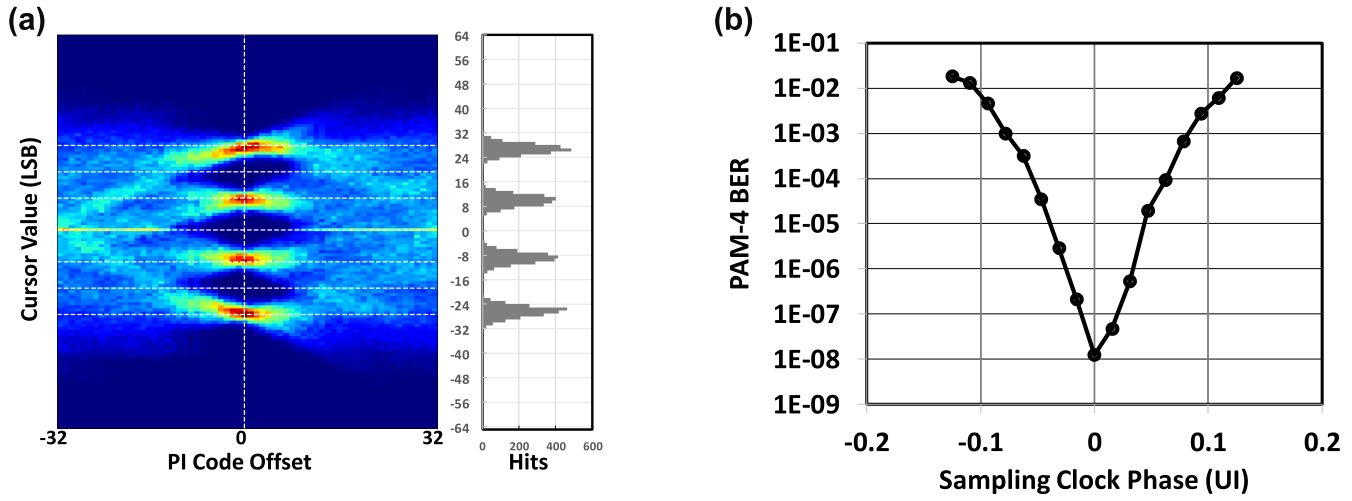


Fig. 15. Receiver performance for 112-Gb/s PRBS-31 over a long-reach channel (37.5 dB). (a) RX eye and cursor histogram. (b) BER bathtub curve.

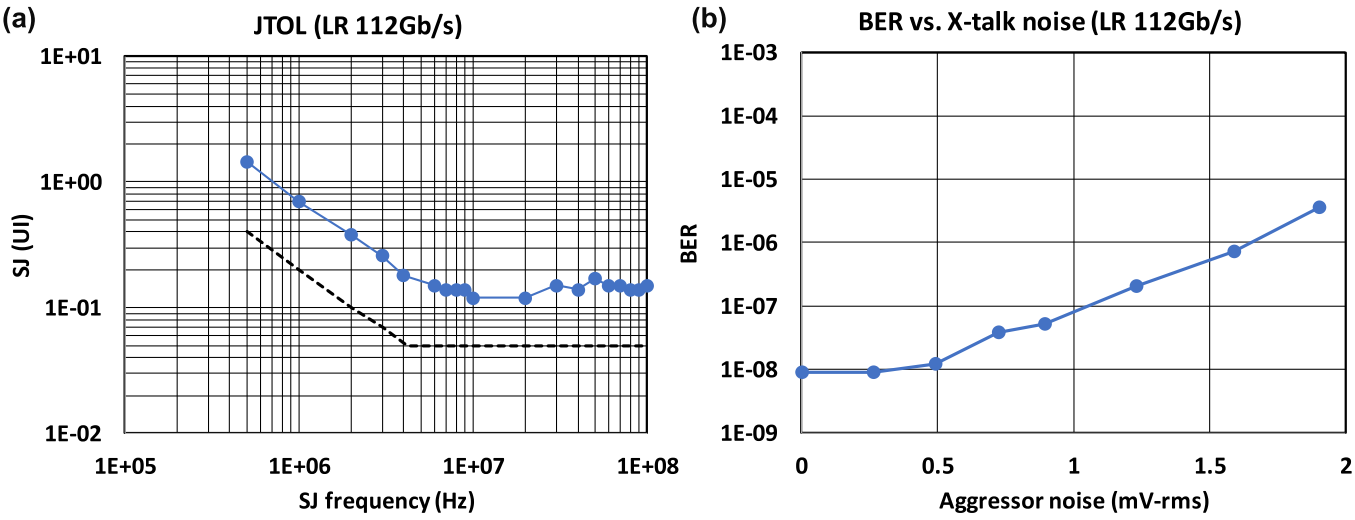


Fig. 16. (a) Sinusoidal jitter tolerance. (b) Cross-talk impact test for 112-Gb/s PRBS-31 over an LR channel.

of 36UI period. To meet the non-overlap timing condition, T/H-2 clocks are then trimmed by 2UI to have a 4UI tracking window as illustrated in the timing diagram in Fig. 11(b).

V. Bench Test

The fabricated transceiver wafer dice are assembled in ball grid array (BGA) packages, following a standard mass production VLSI packaging flow and are tested in a socket mounted on the device under test (DUT) PCB board. A bench test setup for TX-to-RX link performance over a long-reach channel is illustrated in Fig. 12. The DUT board with a DUT package socket and breakout cable connectors is controlled by another general-purpose FPGA board that fetches data and status bits from DUT and sends commands back to DUT. A control software performing PAM-4 RX adaptation and closed-loop corrections is coded and programmed onto the FPGA. An on-chip PRBS checker monitors the DSP data output to measure PAM-4 BER of the sliced data without sub-sampling. Contiguous raw ADC outputs (8-K samples) are captured and stored in the on-chip controller memory upon a trigger signal from the off-chip FPGA controller board. The FPGA controller periodically fetches the stored ADC output samples to perform continuous adaptation of data/error slice levels, FFE/DFE coefficients, CTLE peaking codes, PGA gain codes, AFE offset and baseline wandering correction DAC code, and ADC calibration codes (offset, gain, and skew). An external reference clock generator drives the on-die LC-PLL in DUT.

Fig. 13(a) and (b) shows the 112-Gb/s PAM-4 TX eye diagrams for the PRBS-7 and PRBS-31 patterns, respectively, using the built-in FFE of the Keysight DSA-X 96204Q oscilloscope. The TX output goes through 2 in of Megtron-6 PCB trace and a 24-in-long HUBER+SUHNER cable into the oscilloscope. PRBS-31 eyes show more symmetric PAM-4 eyes, albeit with reduced eye openings. The maximum TX swing is 1-V differential peak-to-peak and the measured random jitter is 140-fs rms for a clock pattern.

Fig. 14(a) and (b) plots the comparison between the measured and simulated composite transfer functions (TFs) of CTLE stages and PGA stages, respectively. The dashed lines

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with markers represent the silicon-measured TFs for various peaking or gain code combinations. The solid lines represent the corresponding simulations. The TF mismatch between the measured data and simulation results is less than 0.5 dB across dc to 30 GHz, supporting the accuracy of our simulation methodology (e.g., parasitic extraction and field solver settings) for inverter-based CTLE and PGA circuits over a broad frequency range. The measured CTLE TF shows a max peaking of 17.5 dB at 30 GHz and its low-frequency zero is located near 300 MHz for the settings used. The measured PGA gain tuning range of 10.5 dB is observed.

To test the link performance, the TX output from the DUT board is looped back to the RX input via a channel consisting of TX/RX package traces (>20 mm total length embedded in an organic laminate substrate), two 1.5-in PCB traces, four SMK connectors, a dc-blocking cap, a pair of 12-in HUBER+SUHNER cables, two break-out boards, and a 0.8-m cable (Fig. 12). The total IL from TX to RX at 28 GHz is measured at 37.5-dB die-to-die, or 33.5-dB BGA-to-BGA excluding device package and DUT board socket, as illustrated in the SDD21 profile in Fig. 12. The measured transceiver performance is summarized as shown in Fig. 15(a) and (b). The transmitted 112-Gb/s PRBS-31 PAM-4 signal over the channel is effectively equalized by the RX at a BER <= 1E-8, as reported by the on-chip PRBS checker. The TX is configured to send a 930-mVppd signal with ∼2-dB pre-cursor and ∼5-dB post-cursor FFE boost.

The receiver eye in Fig 15(a) is plotted by accumulating 8-K symbol values of the equalized data slicer input in DSP while sweeping the sampling phase by overriding the PI code register by −32 to +32 offset around the CDR lock position. The corresponding cursor histogram is a vertical (ADC values measured in the unit of LSB) cross section of the RX eye taken at the PI code offset set to zero (i.e., at the CDR locking point), clearly showing equalized PAM-4 symbol levels. The measured horizontal BER bathtub in Fig. 15(b) shows a margin of >0.1UI to satisfy the OIF CEI pre-forward error correction (FEC) spec [6] of 1E-4 around the CDR locking point, where the BER reaches 1E-8 or lower.

A sinusoidal jitter (SJ) tolerance (JTOL) test in Fig. 16(a) was performed by setting a bit error rate tester (BERT) to inject SJ of varying frequency and magnitude to the DUT receiver over an LR channel. The data show an eye opening margin larger than 0.1UI at high frequency, while satisfying the OIF CEI JTOL mask [6] shown in the dashed line with a good margin across the frequency range from 500 kHz to 100 MHz. The results demonstrate the effectiveness of ISI cancellation by CTLE, FFE, and DFE as well as the jitter filtering capability of the on-chip digital CDR loop.

The cross-talk (X-talk) performance of ultra-high-speed SERDES, such as 112-Gb/s PAM-4 transceiver as reported in this article, has become very important due to the stronger aggressor-to-victim coupling resulting from sharper transition edges and the reduced SNR margin in PAM-4 signaling. The silicon characterization data of X-talk performance of such links provide valuable information to assist and guide system-level link margin budgeting and component specification. For X-talk noise margin testing, an asynchronous 56-Gb/s NRZ

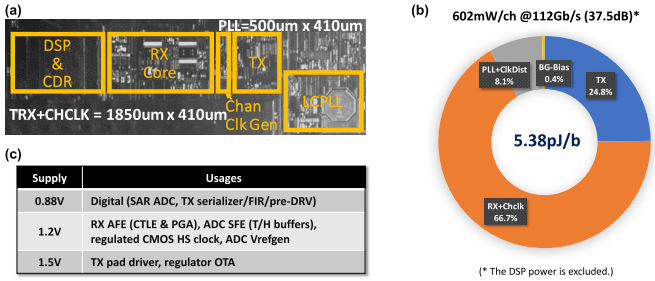


Fig. 17. (a) Die micrograph. (b) Power break-down. (c) Supply usages.

	This work	Ref [1]	Ref [2][4]	Ref [5]	Ref [3]
Technology	7nm FinFET	10nm CMOS	16nm FinFET	14nm FinFET	10nm FinFET
Data Rate	112Gb/s	112Gb/s	112Gb/s	128Gb/s	112Gb/s
TX Architecture	Quarter-rate CML 4-Tap FFE	-	Quarter-rate SST 4-Tap FFE	Quarter-rate CML 3-Tap FFE	Quarter-rate CML 3-Tap FFE
RX Equalization	CTLE 31-Tap FFE 1-Tap DFE	CTLE 16-Tap FFE 1-Tap DFE	CTLE 31-Tap FFE 1-Tap DFE	-	-
Active Area (excl. DSP)	TX: 0.136mm² RX: 0.265mm²	RX: 0.281mm²	TX: 0.38mm² RX: 0.674mm²	TX: 0.048mm²	TX: 0.03mm²
TX Power (excl. PLL)	1.34pJ/b	-	3.08pJ/b	1.0pJ/b (112Gb/s)	1.72pJ/b
RX Power (excl. DSP)	ADC: 4.8pJ/S RX: 3.6pJ/b*	ADC: 6.71pJ/S RX: 4.2pJ/b	ADC: 8.48pJ/S RX: 5.27pJ/b	-	-
PLL and Clk Distribution	0.44pJ/b	-	1.47pJ/b	-	-
Supplies	0.88V/1.2V/1.5V	0.95V/1.5V	0.9V/1.2V/1.8V	0.85V/0.9V	1.0V
Channel	37.5dB	35dB (BERT)	20dB (BERT)	-	-
BER (PRBS31)	1E-8	1E-6	2E-5	-	-

Fig. 18. Performance summary and comparison to prior published 112-Gb/s PAM-4 RX and TX with integrated PLL.

signal of varying amplitude from a BERT is injected into a neighboring channel as an aggressor. The total integrated noise power (in mV-rms) applied at the victim DUT RX input is first calibrated using a scope to measure the RMS difference of noise amplitude with and without the aggressor turned on. The IL between the scope and the X-talk test points was independently measured and accounted for. Plotted in Fig. 16(b), the BER degrades from ∼1E-8 to ∼1E-5 as the aggressor noise increases from 0- to 2-mV rms, leaving 10× margin from the OIF CEI pre-FEC BER spec of 1E-4.

Fig. 17(a) shows a die micrograph of the fabricated chip obtained by the back-side infrared imaging technique with major sub-blocks of the transceiver and the shared LC-PLL demarked by boxes. The measured power of 602 mW per transceiver channel for an LR channel shown in Fig. 17(b) consists of 25% of TX, 66% of RX, and 8% of amortized PLL and high-speed clock distribution contributions. A sub-ADC operating at 1.56 GS/s consumes 2.72 mW of active power, including T/H-2 clock buffers, a StrongArm slicer, SAR logic, CDAC, reference voltage, and ADC output retimers. Three power domains are used throughout the transceiver: 0.88 V for CMOS digital blocks; 1.2 V for RX AFE and ADC SFE stages and regulated supplies for LC-PLL, CMOS clock buffers, and ADC reference voltage; and 1.5 V for regulator error amplifiers and CML TX pad driver front-end.

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VI. Conclusion

Fig. 18 summarizes the transceiver performance and its comparison with prior published 112-Gb/s transmitters and receivers. The entire TX and RX dissipate 1.34 and 3.6 pJ/b, respectively, at 112 Gb/s over a long-reach channel, excluding DSP but including shared channel clock blocks. PLL and clock distribution buffers add 0.44 pJ/b per channel. The active TX and RX (excluding DSP and CDR) area are 0.136 and 0.265 mm^2, respectively.

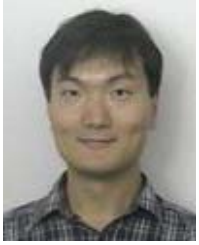
Compared with our previously reported 16-nm 112-Gb/s TX and RX [2], [4], this work achieved 60% of area and 40% of power reduction, thanks to the area-efficient all CMOS receiver front-end and clock generation and distribution circuits as well as a compact 36-way time-interleaving ADC SFE architecture. The bench test results demonstrated a superior performance over long-reach channels, satisfying the BER and JTOL requirements of OIF CEI-112G-LR-PAM4 standard [6].

Acknowledgment

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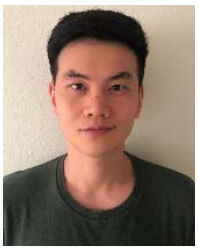
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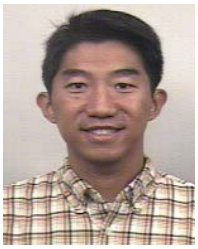
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